

Empirical Characterization of Rationale Stability Under Controlled Perturbations for Explainable Pattern Recognition

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SCAN FOR CODE

THE PROBLEM

XAI evaluation is overwhelmingly **instance-centric**. A method is judged by whether one explanation looks plausible for one input. Nothing in that procedure asks the question a deployed system actually raises.

Does the model reason the same way about inputs it treats the same way?

If an attribution map reshuffles when a sentence is reworded but the prediction holds, the explanation is describing explainer noise as much as model behaviour. In high-stakes settings, that gap is where trust quietly fails.

CONTRIBUTION

- A metric for rationale stability.** ESS scores whether attribution patterns hold across same-label inputs, not whether one explanation looks reasonable.
- A full evaluation grid.** Three transformer encoders, two corpora, both splits, benchmarked against masking-based fidelity.
- A diagnostic use.** Paraphrase ablation exposes rationale drift that leaves the predicted label untouched.

WHAT ESS ADDS

Fidelity asks

Do the highlighted tokens matter?
Mask them and watch the probability fall.

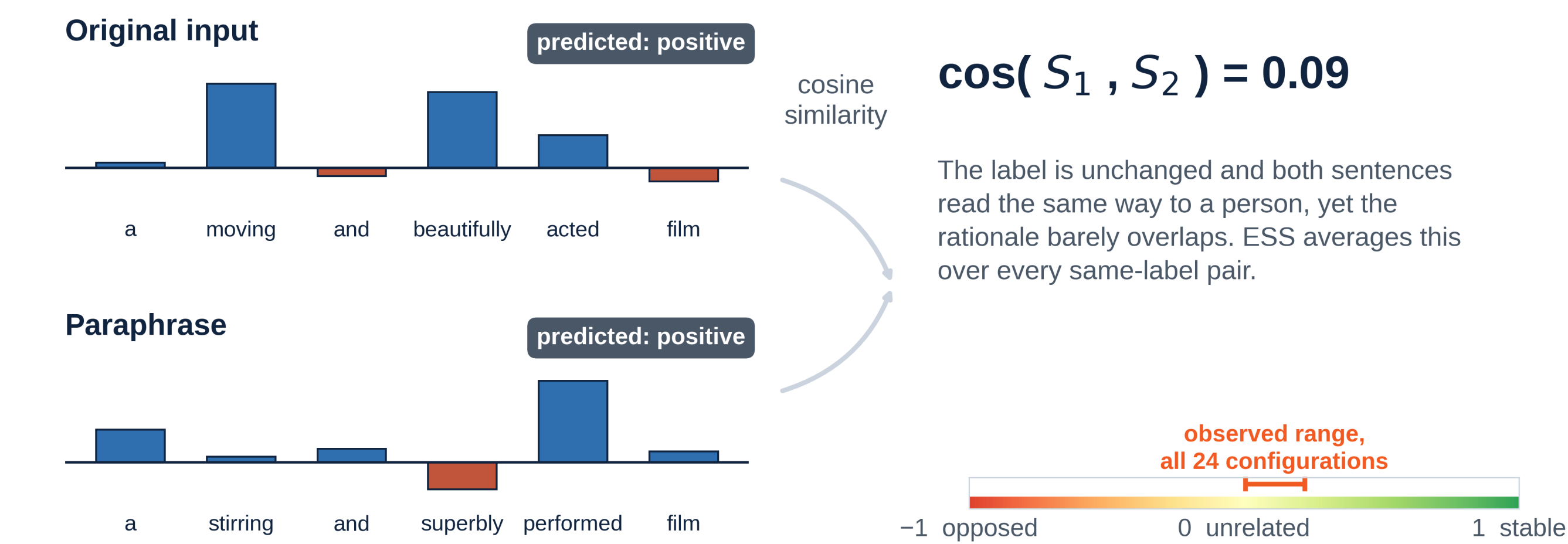
Robustness asks

Does the prediction survive an edit?
It watches the label, not the reason.

ESS asks

Does the reasoning survive an edit? It compares attribution vectors across same-label pairs.

THE METRIC



Illustrative pair. Both are classified positive, yet their SHAP vectors barely overlap.

Explanation Stability Score. Mean cosine similarity of SHAP vectors over all same-label pairs:

$$ESS = \frac{1}{N} \sum_{i=1}^N \frac{S(x_1^{(i)}) \cdot S(x_2^{(i)})}{\|S(x_1^{(i)})\| \|S(x_2^{(i)})\|}$$

ESS $\rightarrow$ 1: same rationale for the same class. ESS $\rightarrow$ 0: unrelated. ESS $<$ 0: opposed.

Fidelity at k . Mask the k highest-|SHAP| tokens and measure the confidence drop:

$$FID_k = \frac{1}{M} \sum_{i=1}^M (p_i - p'_i)$$

Fidelity asks whether the highlighted tokens matter. ESS asks whether the same tokens get highlighted twice.

SETUP

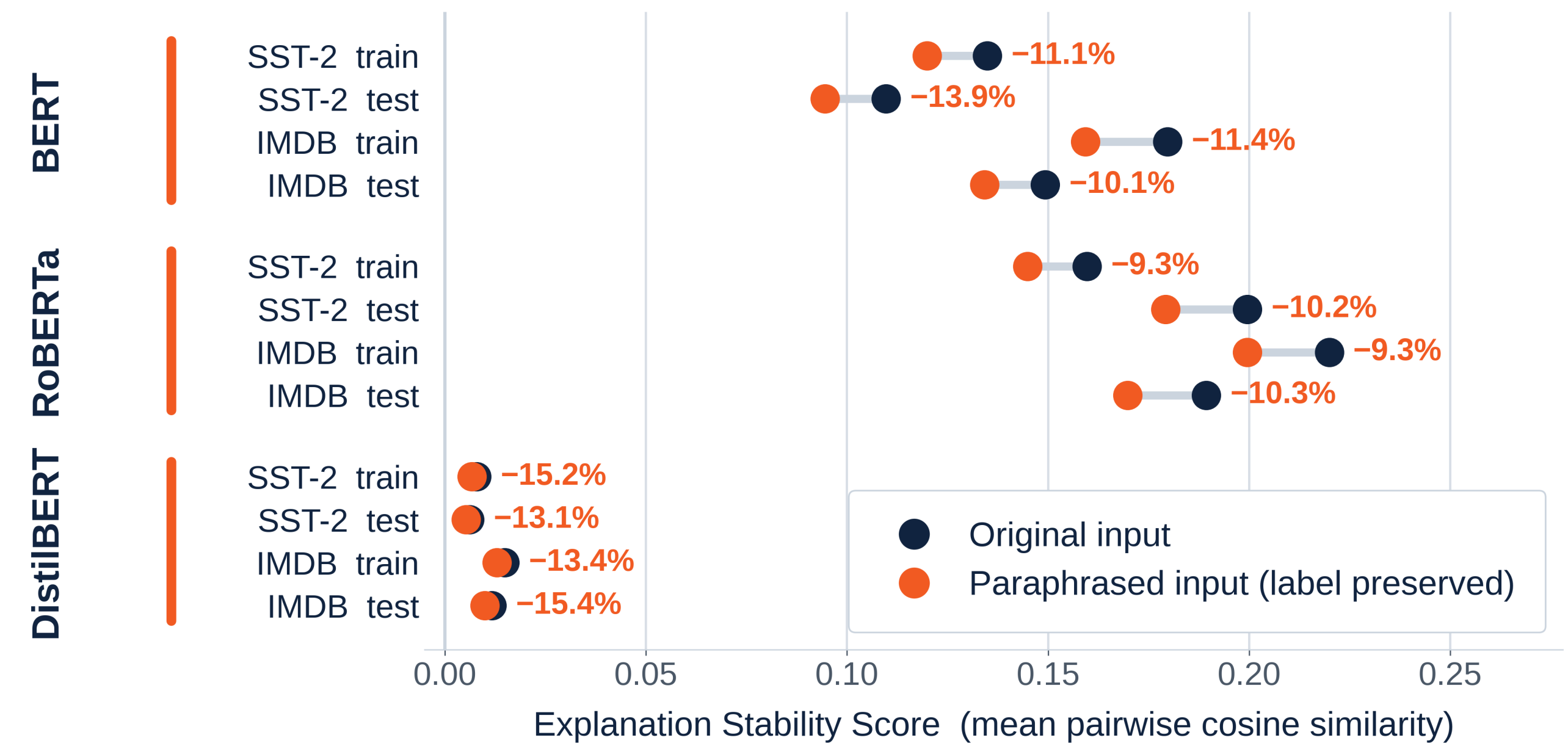
Models	BERT-base-uncased (110M), RoBERTa-base (125M), DistilBERT-base-uncased (66M)
Data	SST-2 and IMDB, train and test splits
Attribution	SHAP KernelExplainer, 100 evaluations per sample, max length 512
Perturbation	WordNet synonym substitution, 20% replacement probability, label preserved
Compute	6 $\times$ NVIDIA RTX 6000, PyTorch 2.3.0; seeds fixed at 42

RESULTS

Model	Data	Split	ESS orig. $\uparrow$	ESS para. $\uparrow$	Δ	FID ₅ $\uparrow$
BERT	SST-2	train	0.1349	0.1199	-11.1%	0.0152
		test	0.1097	0.0945	-13.9%	0.0199
	IMDB	train	0.1797	0.1593	-11.4%	0.0249
		test	0.1493	0.1342	-10.1%	0.0217
RoBERTa	SST-2	train	0.1597	0.1449	-9.3%	0.0021
		test	0.1995	0.1792	-10.2%	0.0025
	IMDB	train	0.2199	0.1995	-9.3%	0.0022
		test	0.1893	0.1698	-10.3%	0.0021
DistilBERT	SST-2	train	0.0079	0.0067	-15.2%	0.0041
		test	0.0061	0.0053	-13.1%	0.0052
	IMDB	train	0.0149	0.0129	-13.4%	0.0057
		test	0.0117	0.0099	-15.4%	0.0053

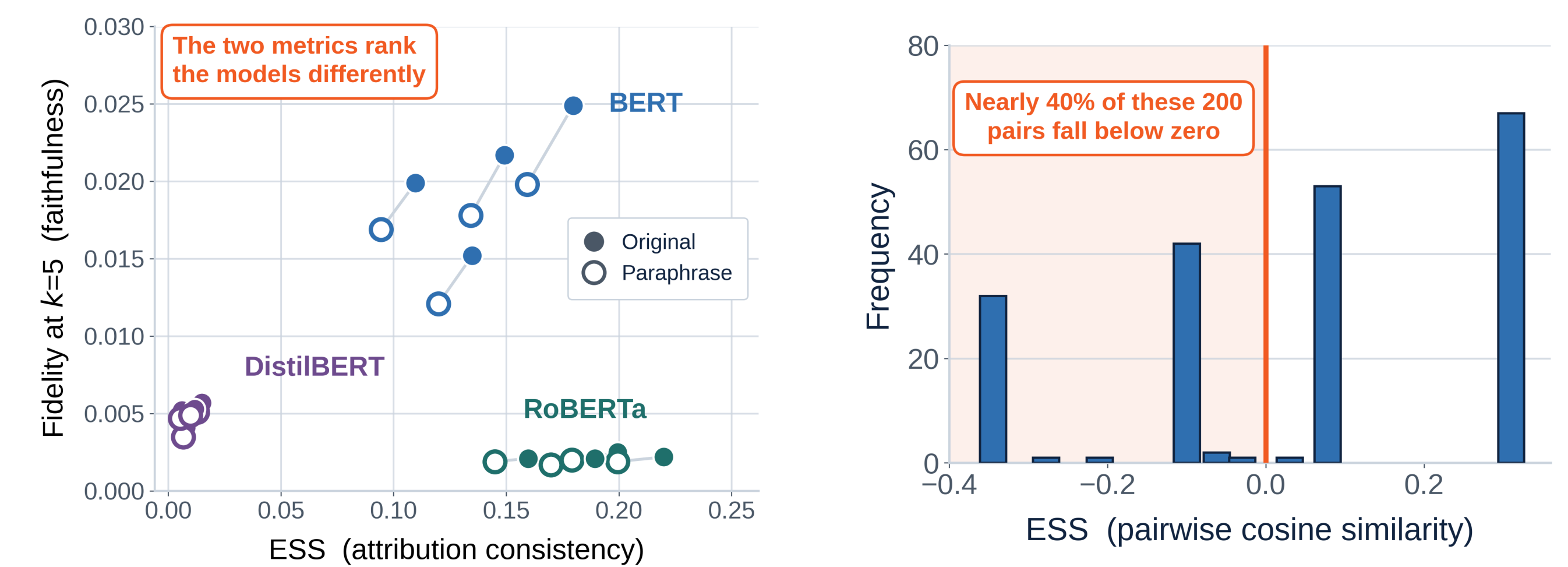
Δ is the relative change in ESS after paraphrase, negative in all twelve configurations. t-test on BERT / SST-2 test, 200 samples: t = 2.32, p = 0.021.

EVERY CONFIGURATION LOSES STABILITY



Original against paraphrased ESS for all twelve configurations, redrawn from the table above. The shift is downward without exception and the ordering between models is preserved, so the effect is not an artefact of any one dataset or split.

CONSISTENCY IS NOT FAITHFULNESS



Left: RoBERTa is the most consistent model and the least faithful, DistilBERT the reverse, so the two metrics order the models differently. The paper notes RoBERTa's low fidelity may itself be a scaling artefact. Right: the pairwise ESS distribution for one sampled set of 200 same-label pairs is multi-modal, and a large mass sits below zero where attributions point in opposing directions.

READING

- Attribution stability is low everywhere.** The best of twelve configurations reaches ESS = 0.22, and in the sampled distribution nearly 40% of pairs fall below zero.
- Label-preserving edits still move the rationale.** A 20% synonym swap costs 9-15% of ESS in every setting while the prediction stands.
- Consistency and faithfulness rank models differently.** RoBERTa is the most consistent and the least faithful, so one score cannot stand in for the other.
- Capacity is not the whole story.** DistilBERT's near-zero ESS reflects flatter, less differentiated attributions. Low variance is not stability.
- ESS is a proxy, not ground truth.** It measures a fixed explainer over a fixed model, so it inherits SHAP's own sampling noise.

CONCLUSION

A stable prediction is not a stable rationale. Across three encoders and two corpora, attributions for same-label inputs agree weakly at best, and every label-preserving paraphrase degrades them further. ESS costs one cosine per pair, captures something fidelity does not, and belongs beside it in any XAI evaluation. Future work will build consistency into training rather than checking for it after the fact.

0.22

Best ESS observed

9-15%

ESS lost to paraphrase

12 / 12

Configurations that lose ESS

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